PHASE 0 - DESIGN DOCUMENT

1. Mathematical Problem Formulation

Let D_t denote warehouse demand during hour t. The task is one-step-ahead forecasting: use the previous 24 hourly observations to predict demand in the next hour.

$$X_t = [D_{t-23}, D_{t-22}, ..., D_{t-1}, D_t], \quad y_t = D_{t+1}.$$

The 24-hour window represents one complete daily cycle. This gives the model enough temporal context to observe within-day structure while keeping the forecasting problem small enough to interpret.

The implemented model first maps the scalar demand value at each time step into a d_model-dimensional representation. It then adds sinusoidal positional information, applies one self-attention head, and uses the representation at the final time position for the next-hour prediction.

$$Q = X W_Q, \quad K = X W_K, \quad V = X W_V.$$

The attention score matrix is computed as the query-key dot product. The scores are divided by sqrt(d_k) before softmax. This is the scaled dot-product attention rule implemented directly in code.

$$S = Q K^T, \quad S_{scaled} = S / \sqrt{d_k}.$$

$$A_{ij} = \exp(S_{scaled,ij}) / \sum_j \exp(S_{scaled,ij}).$$

For numerical stability, the implementation subtracts the largest score in each row before exponentiation. The attention output is then H = A V. No high-level Transformer or attention API is used.

$$PE(pos, 2i) = \sin(pos / 10000^{(2i/d_{model})})$$

$$PE(pos, 2i+1) = \cos(pos / 10000^{(2i/d_{model})}).$$

Positional encoding is included because self-attention by itself does not encode the order of observations. Here, order matters because D_t and D_(t-23) have different forecasting meaning even when their numerical values are similar.

$$MSE = (1/N) * \sum_i (y_i - \hat{y}_i)^2.$$

Mean Squared Error (MSE) is the training objective because it gives larger penalty to large forecast misses. That is useful in this experiment because rare spikes can create operationally important errors.

2. Synthetic Dataset Design

The synthetic environment is designed specifically for the warehouse forecasting problem and is frozen before model evaluation. It contains 365 days of hourly observations, giving 8,760 hourly demand values. The primary random seed is 42.

$$D_t = \max(0, B + T_t + S_t + W_t + E_t + \text{epsilon}_t).$$

The max(0, ...) operation enforces the physical constraint that warehouse demand cannot be negative.

Baseline level: B = 300 units. This sets the normal operating scale of the warehouse.

$$T_t = 0.005\, t.$$

Trend is intentionally small. It makes the series slowly increase over the year, so the process is not perfectly stationary.

$$S_t = 45 \sin(2*\text{pi}*h_t/24) + 20 \cos(2*\text{pi}*h_t/24).$$

The daily seasonal component uses both sine and cosine terms. Together they create a smooth 24-hour cycle with a phase that is not forced to begin exactly at a sine maximum or minimum.

Weekly effects are Monday +10, Tuesday +15, Wednesday +12, Thursday +18, Friday +25, Saturday -15, and Sunday -25. This creates a second temporal pattern beyond the daily cycle.

$$\text{epsilon}_t \sim \text{Normal}(0, 12^2).$$

The ordinary noise term makes the problem stochastic. Without this noise, the experiment would be too close to reproducing a deterministic formula.

Positive demand spikes occur with probability 0.005 per hour and have a Uniform(100, 220) magnitude. Negative demand drops occur with probability 0.003 per hour and have a Uniform(80, 140) magnitude.

These rare events are deliberately included as hard cases. The next hour can change sharply even though the preceding 24 hours do not contain an explicit event indicator.

3. Statistical Assumptions

The design uses the following explicit assumptions:

1. Additive structure. Baseline, trend, daily seasonality, weekly effects, ordinary noise, and rare events are added together before applying the non-negativity constraint. This keeps each source of variation interpretable.

2. Zero-mean ordinary noise. The regular noise has expected value zero, so it does not systematically raise or lower demand.

3. Independent ordinary noise across hours. Predictable temporal dependence therefore comes mainly from the designed trend and seasonal components rather than from autocorrelated random noise.

4. Fixed rare-event probabilities. In the base environment, the probability of a positive spike is 0.005 per hour and the probability of a negative drop is 0.003 per hour.

5. Non-negative demand. Values below zero are clipped to zero because negative warehouse demand has no physical meaning.

6. Information constraint. The forecasting model receives only the previous 24 observed demand values. No future values or external event indicators are supplied.

7. Train-only preprocessing. When standardization is used, the mean and standard deviation are fitted on the training data only and then applied unchanged to validation and test data. This prevents temporal information leakage.

4. Explicit Hypotheses

Hypothesis 1 - Temporal weighting: because the data contain daily and weekly structure, self-attention should learn non-uniform temporal weights across the 24-hour context rather than treating every lag as equally informative.

Hypothesis 2 - Forecasting value: the attention model should improve at least some forecasting error characteristics relative to simple forecasting rules because it can combine information from multiple positions in the 24-hour context.

Hypothesis 3 - Robustness: a fixed model should lose accuracy under distribution shifts that change the demand level or substantially increase noise because the learned mapping was fitted to the original environment.

5. System-Specific Failure Modes

Failure mode 1 - Rare spikes or drops. A sudden positive event of roughly 100-220 units or negative event of roughly 80-140 units may occur without a dedicated predictor in the input. The model can therefore produce a large error.

Failure mode 2 - Demand-level shift. A +40 unit shift in the test environment can create systematic forecast bias because the model was trained on the original demand level.

Failure mode 3 - Increased noise. Doubling the ordinary noise standard deviation from 12 to 24 lowers the signal-to-noise ratio and makes next-hour demand less predictable from recent history.

Failure mode 4 - Over-reliance on persistence. Because consecutive hourly observations are correlated, the model could obtain good average performance mainly by emphasizing the latest value rather than learning broader context. This is investigated with attention-by-lag analysis and baseline comparison.

Failure mode 5 - Numerical instability. Exponentiating large attention scores can overflow in a naive softmax implementation. The implemented softmax subtracts the row maximum before exponentiation, and a separate numerical-stability test checks this behavior.

6. Evaluation Methodology

The 8,760-hour series is converted into supervised examples using a 24-hour input window and a one-hour-ahead target. There are 8,736 valid supervised examples.

The examples are divided chronologically by target timestamp into 70% training, 15% validation, and 15% test. This is essential for a time-series experiment because a random split could allow future observations to influence training.

A validation example near a split boundary can use the immediately preceding observed hours from the earlier period. This is intentional because those observations would genuinely be available at forecast time. The split is by target timestamp rather than by discarding historical context.

$$MAE = (1/N) * sum_i |y_i - yhat_i|.$$

Mean Absolute Error (MAE) measures the typical absolute forecasting error in demand units.

$$RMSE = sqrt((1/N) * sum_i (y_i - yhat_i)^2).$$

Root Mean Squared Error (RMSE) gives more influence to large misses. This is particularly useful for identifying failures around spikes and drops.

$$MAPE = (100/N) * sum_i |(y_i - yhat_i) / y_i|.$$

Mean Absolute Percentage Error (MAPE) expresses the error relative to the actual demand value, making the result easier to interpret across demand levels.

Three explicit baselines are evaluated on the same test targets: persistence, previous-day same-hour, and 24-hour mean.

Persistence predicts D_(t+1) using D_t. It measures how much accuracy can be obtained from immediate temporal continuity alone.

Previous-day same-hour predicts D_(t+1) using D_(t-23). It is designed to capture the daily cycle in the synthetic data.

The 24-hour mean predicts the next hour using the mean of the input window. It is a smoothing baseline that represents a typical recent level.

The learned attention model uses d_model = 4, one attention head, the direct attention implementation, sinusoidal positional encoding, Adam with learning rate 0.01, MSE loss, 200 epochs, and seed 42. The checkpoint with the best validation loss is restored before final test evaluation.

The trained model is also evaluated under controlled distribution shifts without retraining. Finally, the largest absolute forecast errors are inspected together with target jumps and average final-query attention by lag.

7. Expected Baseline Behavior

Persistence is expected to be strong because consecutive hourly observations are temporally correlated. It should perform especially well during smooth periods but cannot anticipate an unexpected spike or drop.

Previous-day same-hour is expected to capture the explicit 24-hour seasonality. It can become less accurate when trend, noise, weekly effects, or rare events make the current day different from the previous day.

The 24-hour mean is expected to smooth noise, but it should lag behind local peaks, troughs, and sudden changes because it ignores the exact current state.

The attention model is expected to combine short-term continuity with information from other points in the 24-hour context. The design does not assume it must outperform every baseline on every metric; the purpose of the baselines is to establish interpretable reference behavior.

8. Planned Generalization Experiment

Generalization is tested by freezing the trained attention model and changing the test data-generating environment without retraining. This isolates transfer to changed conditions from adaptation after retraining.

Experiment A - Level shift. Add +40 units to the demand level. This tests sensitivity to a changed mean level while retaining the underlying temporal structure.

Experiment B - Noise shift. Increase ordinary noise standard deviation from 12 to 24. This tests robustness when observations contain substantially more unpredictable variation.

For each shift, persistence, previous-day same-hour, 24-hour mean, and the frozen attention model are evaluated using MAE, RMSE, and MAPE.

The key questions are how much error changes under each shift, whether the learned model remains useful relative to simple rules, and whether the observed degradation is consistent with the mechanism of the shift.

Phase 0 conclusion: The design is the reference specification for the later mathematics, learning experiment, warehouse forecasting, distribution-shift analysis, failure investigation, and reproducibility checks. Any result in the later documents should be interpreted against these pre-defined assumptions and hypotheses.